House Price Prediction - Summary Report

Objective

The goal of this project is to predict house prices using the Ames Housing Dataset and compare the performance of different regression algorithms.

Data Preprocessing

- Handled missing values in numerical and categorical features.
- Removed outliers using the IQR method.
- Converted categorical variables into numerical format using one-hot encoding.

Models Used

1. Linear Regression
2. Ridge Regression
3. Random Forest Regression

Evaluation Metrics

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Results

Model	MAE	RMSE	R^2 Score
Linear Regression	10562	14328	0.9384
Ridge Regression	10452	14401	0.9378
Random Forest	11862	16890	0.9144

Conclusion

Linear Regression achieved the best overall performance with the highest R^2 score and lowest prediction error. The model successfully predicts house prices and demonstrates the effectiveness of regression techniques for real estate price estimation.